VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY
SUBJECT-- MATRICES AND CALCULUS
WORK SHEET PROBLEMS FOR TUTORIAL CLASS
MODULE -I
MATRICES

SHORT ANSWER QUESTIONS

1. Define (a) Rank of a Matrix (b) Echelon form of a Matrix
2. Define the Normal Form of a Matrix
3. Define the Inverse of a Matrix
4. Discuss the nature of the solution of the system of Non-Homogeneous Linear Equations AX=B.
5. Discuss the nature of the solution of the system of Homogeneous Linear Equations AX=O.
6. Find the value of k so that the rank of the matrix $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & k & 0 \end{bmatrix}$ is 2
7. Find the value of k such that the rank of a matrix A = $\begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$ is 2
8. Find the rank of the matrix A = $\begin{bmatrix} 1 & -4 & 6 \\ 2 & 3 & 2 \\ 0 & -6 & -5 \end{bmatrix}$ by reducing to echelon form.
9. Find the rank of the matrix A = $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ by reducing to Normal form.
10. What do you mean by a Diagonally Dominant system

LONG ANSWER QUESTIONS

ECHELON FORM OF A MATRIX

1. Find the rank of the matrix A = $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 4 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$ by reducing to echelon form.
2. Find the Rank of the matrix A= $\begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 2 & 3 \\ 2 & 3 & 3 & 3 \end{bmatrix}$ by reducing to echelon form.
3. For what values of a and b is the rank of the matrix A = $\begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & 1 & -1 & 2 \\ 6 & -2 & a & b \end{bmatrix}$ is (i) 2 and (ii) 3
4. Find the Rank of the matrix A= $\begin{bmatrix} 0 & -3 & -4 & 2 \\ -1 & -2 & -5 & -3 \\ 2 & -3 & -6 & -7 \end{bmatrix}$ by reducing to echelon form.
5. Determine the values of b such that the rank of matrix A= $\begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ b & 2 & 2 & 2 \\ 9 & 9 & b & 3 \end{bmatrix}$ is 3

NORMAL FORM OF A MATRIX

1. For values of k, find the rank of the matrix $A=\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & k \end{bmatrix}$ by reducing to Normal form
2. Find the rank of the matrix $A = \begin{bmatrix} 4 & 4 & -3 & 1 \\ 1 & 1 & -1 & 0 \\ -3 & 2 & 2 & 2 \\ 9 & 9 & 4 & 3 \end{bmatrix}$ by reducing to Normal form.
3. Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{bmatrix}$ by reducing to Normal form.
4. Find the rank of the matrix $A=\begin{bmatrix} 3 & 2 & -1 & 5 \\ 5 & 1 & 4 & -2 \\ 1 & -4 & 11 & -19 \end{bmatrix}$ by reducing to Normal form.
5. Find the rank of the matrix $A=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 2 \\ 3 & 3 & 4 \end{bmatrix}$ by reducing to Normal form.

INVERSE OF A MATRIX (GAUSS-JORDAN METHOD)

1. Find the inverse of $A=\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -3 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ by using Elementary Row Transformations

   Verify that $AA^{-1} = I$
2. Find the inverse of the matrix $A= \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix}$ by using the Gauss-Jordan method.

   Verify that $AA^{-1} = I$
3. Find the inverse of the matrix $A= \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ by using Elementary Row Transformations.

   Verify that $AA^{-1} = I$
4. Find the inverse of the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ by using the Gauss-Jordan method.

   Verify that $AA^{-1} = I$

SYSTEM NON-HOMOGENEOUS LINEAR EQUATIONS

1. For what values of a and b, the equations x+2y+3z=8, 2x+y+3z=13, 3x+4y-az=b has (i)No solution, (ii)A Unique Solution, (iii) An Infinite Number of Solutions?
2. For what values of λ the set of equations 2x-3y+6z-5t = 3, y-4z+t = 1 and 4x – 5y +8z -9t = λ has (i) No solution (ii) an infinite number of solutions, and find a solution of the equations when they are Consistent.
3. Determine the values of μ for which the following equations are consistent. Also, solve the system for these values of μ . x+2y+z=3, x+y+z=μ , 3x+y+3z=μ².
4. Solve 4x-y+6z=16, x-4y-3z=-16, 2x+7y+12z = 48, 5x-5y+3z = 0
5. Solve x+2y-z=3, 3x-y+2z=1, 2x-2y+3z=2, x-y+z = -1

6. Solve the system for x, y, z; $\frac{-1}{x} + \frac{3}{y} + \frac{4}{z} = 30, \frac{3}{x} + \frac{2}{y} - \frac{1}{z} = 9, \frac{2}{x} - \frac{1}{y} + \frac{2}{z} = 10.$
7. Find the values of a and b for which the equations x+ ay+ z =3, x+2y+2z=b, x+5y+3z=9 are consistent. When will these equations have infinitely many solutions?
8. Show that the equations -2x+y+z=a, x-2y+z=b, x+y-2z=c have no solution unless a+b+c=0, in which case they have infinitely many solutions. Find these solutions when a=1, b=1, c=-2

SYSTEM HOMOGENEOUS LINEAR EQUATIONS

1. Show that the only real number k for which the system x+2y+3z=kx, 3x+y+2z=ky, 2x+3y+z=kz has a non-zero solution is 6, and solve them when k=6.
2. Solve the system of equations kx+y+z=0, x+ky+z=0, x+y+kz = 0 for all values of k,
3. Solve for all values of k the system of equations 2x+3ky+(3k+4) z=0, x+(k+4) y+(4k+2) z=0, x+2(k+1) y+(3k+4) z=0.
4. Solve x+y-3z+2w=0, 2x-y -2z-3w = 0, 3x-5y-w = 0, 5x -y-7z -4w = 0
5. For what real values of k, does the system of equations possess a non-trivial solution 3x+y-kz=0, 4x+2y-3z=0, 2kx+4y-kz=0 and solve the system for that k value.
6. Show that the system of equations 2x-2y+z = kx,2x-3y+2z = ky, -x+2y = kz can possess a non-trivial solution only if k=1, k=-3. Obtain the general solution in each case.

GAUSS SEIDEL ITERATION METHOD

1. Explain the Gauss-Seidel Method.
2. Solve using Gauss -Seidel Method 5x-y+z =10, 2x+4y =12, x+y+5z = -1 start with the solution (1.5, 2.5, 0).
3. Solve the system 10x+2y+z = 9, 2x+20y-2z = -44, -2x+3y=10z = 22 using the Gauss-Seidel Method.
4. Solve 27x+6y-z = 85, x+y+54z = 110, 6x+15y+2z = 72 using the Gauss-Seidel method.
5. Solve by the Gauss-Seidel iteration method, the equations 20x+y-2z = 17, 3x+20y–z = -18, 2x-3y+20z = 25
6. Solve the system 6x+y+z = 20, x+4y -z = 6, x-y+5z = 7 using Gauss seidel method

MODULE II

EIGEN VALUES , EIGEN VECTORS, AND QUADRATIC FORMS

SHORT ANSWER QUESTIONS

1. Define (a) Eigen Value (ii) Eigen Vector
2. State Cayley Hamilton Theorem
3. Define (a) Modal matrix, (b) Spectral matrix (c)Algebraic multiplicity, (d)Geometric multiplicity
4. Define a Quadratic form. Give an Example
5. Define Nature, Index, Signature and Rank of a quadratic form
6. Find Nature, Index, Signature and Rank of quadratic form 2xy+2yz+2zx
7. Find the vector X orthogonal to the vectors $[1\ 0\ -1]^T$ and $[1\ 2\ 1]^T$
8. If $A=\begin{bmatrix}1 & 2 & -2\\ 1 & 1 & 1\\ 1 & 3 & -1\end{bmatrix}$, find the eigen values (a) A^2 (b) A^{-1} (c) AdjA (d) A-3I (e) $A^2 + 3A+I$. Also find the trace and determinant of the matrix $A^2 + 3A+I$.
9. If $A = \begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$, Using Cayley Hamilton theorem, find A^3
10. If $A = \begin{bmatrix}1 & 2\\ 2 & -1\end{bmatrix}$, Using Cayley Hamilton theorem, find A^{-1}

LONG ANSWER QUESTIONS

1. Given $A=\begin{bmatrix}-1 & 2 & -2\\ 1 & 2 & 1\\ -1 & -1 & 0\end{bmatrix}$ (i) Diagonalise the matrix A Hence, find A^4
   (ii) Verify Cayley Hamilton theorem for the matrix A. Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8- A^7 - 5A^6+5A^5+ A^4 - 5A^3 +8A^2 -2A+I$ as a quadratic polynomial in A, and also find the matrix.
2. Given $A=\begin{bmatrix}3 & 1 & -1\\ -2 & 1 & 2\\ 0 & 1 & 2\end{bmatrix}$ (i) Diagonalize the matrix A Hence, find A^4
   (ii) Verify Cayley Hamilton theorem for the matrix A. Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8- 6A^7 +11A^6 -6A^5+ A^4 - 5A^3 +2A^2 - 2A-I$ as a quadratic polynomial in A, and also find the matrix.
3. Given $A=\begin{bmatrix}5 & -6 & -6\\ -1 & 4 & 2\\ 3 & -6 & -4\end{bmatrix}$ (i) Diagonalize the matrix A. Hence, find A^4.
   (ii) Verify Cayley Hamilton theorem for the matrix A. Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8- 5A^7 +8A^6 –4A^5+ 2A^4 -8A^3 -8A^2 – 3A+2I$ as a quadratic polynomial in A, and also find the matrix.
4. Given $A=\begin{bmatrix}1 & 3 & 3\\ 1 & 4 & 3\\ -1 & 3 & 4\end{bmatrix}$ (i) Diagonalize the matrix A if possible.
   (ii) Verify Cayley Hamilton theorem for the matrix A. Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8- 9A^7 +15A^6 –7A^5+ A^4 - 8A^3 -8A^2 - 2A+I$ as a quadratic polynomial in A and also find the matrix.

5. Given $A = \begin{bmatrix} -3 & -2 & 1 \\ -2 & 0 & 4 \\ -6 & -3 & 5 \end{bmatrix}$ (i) Diagonalize the matrix A Hence, find A^4.
   (ii) Verify Cayley Hamilton theorem for the matrix A. Hence find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8 - 2A^7 - A^6 + 2A^5 + A^4 + A^3 - 2A^2 + 3A + I$ as a quadratic polynomial in A, and also find the matrix.

6. Given $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ (i) Diagonalize the matrix A Hence, find A^4.
   (ii) Verify Cayley Hamilton theorem for the matrix A. Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8 - 7A^7 + 11A^6 - 5A^5 + A^4 - 6A^3 - 8A^2 - 3A + I$ as a quadratic polynomial in A, and also find the matrix.

7. Given $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ (i) Diagonalize the matrix A Hence, find A^4
   (ii) Verify Cayley Hamilton theorem for the matrix A Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, express the polynomial $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ as a quadratic polynomial in A, and also find the matrix.

8. Given $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ (i) Diagonalize the matrix A Hence, find A^4
   (ii) Verify Cayley Hamilton theorem for the matrix A Hence, find A^{-1} and A^4.
   (iii) Using Cayley Hamilton theorem, the polynomial $A^8 + A^7 - 21A^6 - 45A^5 + A^4 - 2A^3 - A^2 + 2A + I$ as a quadratic polynomial in A, and also find the matrix.

9. (a) If λ is an Eigen value of a nonsingular matrix A corresponding to the Eigen vector X then prove that λ^{-1} is an Eigen value of A^{-1} corresponding to the Eigen vector X.
   (b) If λ is an Eigen value of a non-singular matrix A corresponding to the Eigen vector X, then prove that $k\lambda$ is an Eigen value of kA corresponding to the Eigen vector X.
   (c) If λ is an eigen value of A corresponding to the eigen vector X then prove that $\lambda+k$ is an eigen value of A+KI corresponding to the Eigen vector X.

10. (a) If λ is an Eigen value of A corresponding to the Eigen vector X then prove that λ^n is an Eigen value of A^n corresponding to the Eigen vector X.
    (b) If λ is an Eigen value of a non-singular matrix A corresponding to the Eigen vector X then prove that $\frac{|A|}{\lambda}$ is an Eigen value of adjA corresponding to the Eigen vector X.
    (c) Prove that sum of eigen values of a matrix is equals to its trace.

11. Reduce the quadratic form $6x^2 + 3y^2 + 3z^2 - 2yz + 4xz - 4xy$ to canonical form by an orthogonal transformation and hence find its signature, nature, rank and index. Also, write the linear transformation that brings about the canonical form.

12. Reduce the quadratic form $x^2 + 3y^2 + 3z^2 - 2yz$ to canonical form by an orthogonal transformation and hence find its signature, nature, rank and index. Also, write the linear transformation that brings about the canonical form.
13. Reduce the quadratic form $8x^2 + 7y^2 + 3z^2 - 12xy - 8yz + 4zx$ to canonical form by an orthogonal transformation and hence find its signature, nature, rank and index. Also write the linear transformation that brings about the canonical form.
14. Reduce the quadratic form $2xy + 2xz - 2yz$ to canonical form by an orthogonal transformation and hence find its signature, nature, rank and index. Also, write the linear transformation that brings about the canonical form.

MODULE III
SINGLE VARIABLE CALCULUS

SHORT ANSWER QUESTIONS

1. Define (a)Limit of a function (b) Continuity of a function
2. State (a) Rolle's Theorem (b) Lagrange's Mean Value Theorem (c)Cauchy Mean value theorem
3. Explain Geometrical Interpretation of (a) Rolle's Theorem (b) Lagrange's Mean Value Theorem
4. State whether Lagrange's Mean Value Theorem for the function f(x) = $|x|$ in [-3,3]. Justify your answer.
5. State whether Rolle's Theorem for the function f(x) = $|xcosx|$ in [0,$\frac{3\pi}{2}$]. Justify your answer.
6. State whether Rolle's Theorem for the function f(x) = $x^3 + 2x - 3$ in [0,2]. Justify your answer.
7. Write the Maclaurins series expansion of the function f(x)= $Tan^{-1}x$
8. Obtain Taylor series expansion of sinx in terms of x- $\frac{\pi}{4}$
9. Obtain the Maclaruarins series expansion of the function f(x)= log(1-x)
10. Using mean value theorem tanx>x on $0 < x < \pi/2$

LONG ANSWER QUESTIONS
ROLLE'S THEOREM

1. Verify Rolle's theorem for the function f(x) = $x^2(1 - x^2)$ in [0,1]
2. Verify Rolle's theorem for the function f(x)= $\log\left(\frac{x^2+ab}{(x(a+b)}\right)$ in [a,b]
3. Check the applicability of Rolles theorem for the function f(x)=e^x (sinx- cosx) in [$\frac{\pi}{4}$,$\frac{3\pi}{4}$]
4. Verify Rolle's theorem for the function f(x)=sinx$\sqrt{cos2x}$ in [0,$\frac{\pi}{4}$]
5. Find the value of c of Rolle's theorem for f(x)= $(x-a)^m(x-b)^n$ in [a, b] where m,n are positive integers
6. By considering the function (x-2)logx show that the equation xlogx =2-x is satisfied that at least one value of x lying between 1 and 2.
7. It is given that Rolle's theorem holds for the function f(x)=x^3+bx^2+cx,1≤x≤2, at the point x=4/3 Find the values of b and c.
8. Using Rolle's theorem, show that f(x)= $8x^3-6x^2-2x+1$ has a zero between in 0 and 1
9. Prove that the equation $2x^3-3x^2-x+1 = 0$ has at least one root between 1 and 2.
10. Use Rolle's theorem, prove that the equation ax^2 bx = ($\frac{a}{3} + \frac{b}{2}$) has a root between 0 and 1

LAGRANGE'S MEAN VALUE THEOREM

1. Verify Lagrange's Mean Value Theorem for the function $2x^2 - 7x + 10$, [2,5]
2. Verify Lagrange's mean theorem for the function f(x)= cosx in [0,$\frac{\pi}{2}$]
3. Verify Lagrange's Mean Value Theorem for f(x)= $\begin{cases} xsin\left(\frac{1}{x}\right) & if\ x \neq 0 \\ 0 & if\ x = 0 \end{cases}$ in [-1,1]
4. Find the point on the curve y=x(x-1)(x-2), at which the tangent is parallel to the chord joining the points (0,0) and ($\frac{1}{2}$,$\frac{3}{8}$)
5. Prove using Lagrange's Mean Value Theorem |cosb-cosa| ≤ |b-a| for all a,b
6. Show that for any x>0, $1+x< e^x<1+xe^x$
7. Using Lagrange's Mean Value Theorem, prove that $0< \frac{1}{\log(1+x)} - \frac{1}{x} < 1$
8. Using Lagrange's mean value theorem, show that $\frac{1}{b} < \frac{logb}{b-1} < 1$, for b>1

9. Using Lagrange's Mean Value Theorem for 0<a<b, prove that $1-\frac{a}{b} < \log\frac{a}{b} < \frac{b}{a} - 1$ and hence show that $\frac{1}{6} < \log\frac{6}{5} < \frac{1}{5}$
10. If a < b, prove that, $\frac{a-b}{\sqrt{1-b^2}} < \cos^{-1}(b) - \cos^{-1}(a) < \frac{a-b}{\sqrt{1-a^2}}$ using Lagrange's mean value theorem
    Hence deduce that $\frac{\pi}{3} - \frac{1}{5\sqrt{3}} > \cos^{-1}\frac{3}{5} > \frac{\pi}{3} - \frac{1}{8}$
11. If a < b, prove that $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1}(b) - \sin^{-1}(a) < \frac{b-a}{\sqrt{1-b^2}}$ using Lagrange's mean value theorem
    Hence deduce that $\frac{\pi}{6} - \frac{1}{5\sqrt{3}} < \sin^{-1}\frac{3}{5} < \frac{\pi}{6} + \frac{1}{8}$
12. If a<b, prove that $\frac{b-a}{1+b^2} < \tan^{-1}(b) - \tan^{-1}(a) < \frac{b-a}{1+b^2}$ using Lagrange's mean value theorem
    Hence, deduce that (i) $\frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$ (ii) $\frac{\pi}{4} + \frac{1}{5} < \tan^{-1}(2) < \frac{\pi}{4} + \frac{1}{2}$

CAUCHY'S MEAN VALUE THEOREM

1. Prove using Cauchy mean value theorem $|\cos b - \cos a| \leq |b-a|$ for all a,b
2. Prove using Cauchy mean value theorem $|\sin b - \sin a| \leq |b-a|$ for all a,b
3. Find the value of c in Cauchy mean value theorem for $f(x) = \sqrt{x}$ $g(x) = \frac{1}{\sqrt{x}}$ in[1,4]
4. Find the value of Cauchy mean value theorem for $f(x) = e^x$ and $g(x) = e^{-x}$ defined On [a,b], 0<a<b
5. Verify Cauchy mean value theorem for f(x) and $f^1(x)$ in [1,e] given f(x) = logx
6. If $0 \leq x \leq 1$, prove that $\frac{\sqrt{1-x}}{\sqrt{1+x}} < \frac{\log(1+x)}{\sin^{-1}x} < 1$

TAYLOR'S SERIES

1. Expand $f(x) = (1-x)^{5/2}$ upto three terms in powers of x.
2. Expand $e^{x\sin x}$ in powers of x
3. Using Maclaurin's series, expand tanx up to the term containing x^5
4. Expand $\frac{\sin^{-1}x}{\sqrt{1-x^2}}$ in powers of x
5. Show that $\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} +$----------
6. Using Taylor's series obtain the value of $\cos 32^0$ correct to four decimal places

CURVE TRACING

1. Trace the curve $y = a\cosh(\frac{x}{a})$
2. Trace the curve $y^2(a+x) = x^2(3a-x)$
3. Trace the curve $x^2y^2 = a^2(y^2 - x^2)$
4. Trace the curve $y^2(x-a) = x^2(a+x)$
5. Trace the curve $xy^2 = 4a^2(2a-x)$
6. Trace the curve $r = a(1+\cos\theta)$
7. Trace the curve $r = a(1+\cos\theta)$
8. Trace the curve $r^2 = a^2\cos 2\theta$

MODULE IV

PARTIAL DIFFERENTIATION AND ITS APPLICATIONS

SHORT ANSWER QUESTION

1. Define (a) Jacobian (b) Functional dependence
2. If u =$Tan^{-1}(y/x)$ then find u_x at the point (0,-1)
3. If u=x(1-y) and v =xy then find the Jacobian J
4. If $u=x^2-y^2$, v=2xy find $\frac{\partial(x,y)}{\partial(u,v)}$
5. Find the Jacobian of transformation from Cartesian to cylindrical coordinates given by x = r$\cos\theta$, y = r$\sin\theta$ and z=z
6. If x = $\frac{u^2}{v}$, y = $\frac{v^2}{u}$ find $\frac{\partial(u,v)}{\partial(x,y)}$
7. (a) When the function f(x,y) with usual notations of partial derivatives coefficients will have a maximum, a minimum and can't be decided?

   (b) Explain the Procedure to find the maximum and minimum of f(x,y)
8. Examine for continuity at the origin of the function defined by f(x,y)= $\left\{ \begin{array}{r} \frac{x^2}{\sqrt{x^2+y^2}} \; for\; x \neq 0, y \neq 0 \\ for\; x = 0, y = 0 \end{array} \right\}$

   Redefine the function to make it continuous
9. Define (a) Saddle point (ii) Stationary point
10. State Eulers Theorem

LONG ANSWER QUESTIONS
PARTIAL DIFFERENTIATION

1. Check whether $f_{xy}=f_{yx}$ for the function f = $\tan^{-1}(\frac{x}{y})$
2. If f = $\log(x^2+y^2) + \tan^{-1}(\frac{y}{x})$ then prove that $f_{xx} + f_{yy} = 0$
3. If $x^y y^y z^z$ = e show that at x = y = z, $\frac{\partial^2 Z}{\partial x \partial y}$ = $-(x\log ex)^{-1}$
4. If u = f(2x-3y, 3y-4z, 4z-2x) the prove that $\frac{1\partial u}{2\partial x}+\frac{1\partial}{3\partial y}+\frac{1\partial}{4\partial z} = 0$
5. If u = $\tan^{-1}(\frac{2xy}{x^2-y^2})$ prove that $u_{xx} + u_{yy} = 0$
6. If $u = sin^{-1}\left(\frac{x}{y}\right) + tan^{-1}(\frac{y}{x})$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$

JACOBIAN

1. If x+y+z=u , y+z=uv, z=uvw then evaluate $\frac{\partial(x,y,z)}{\partial(u,v,w)}$
2. If u= $x+3y^2-z^3$, v=$4x^2yz$, w= $2z^2$-xy evaluate $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at (1,-1,0)
3. If u=x^2-y^2, v=2xy, where x = r$\cos\theta$, y= r$\sin\theta$, find $\frac{\partial(u,v)}{\partial(r,\theta)}$
4. If ux=yz, vy=zx, wz=xy, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$
5. If x= $e^r \sec\theta$, y= $e^r \tan\theta$ Then prove that $JJ^1 = 1$
6. If u = $\frac{x+y}{x-y}$ v= $\frac{xy}{(x-y)^2}$ verify whether u,v are functionally dependent. If so, find the relation between them.

7. If u= $\frac{x+y}{1-xy}$ and v= $\tan^{-1}(x)+\tan^{-1}(y)$. Verify whether u,v are functionally dependent. If so, find the relation.
8. If u = xy+yz+zx, v= $x^2+y^2+z^2$ and w = x+y+z find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ Verify whether u, v and w are functionally dependent. If so, find the relation.
9. Prove that u = $\frac{x^2-y^2}{x^2+y^2}$ v = $\frac{2xy}{x^2+y^2}$ are functionally dependent and find the relation between them.
10. Prove that u=$\sin^{-1}x+\sin^{-1}y$ and v=$x\sqrt{1-y^2}+y\sqrt{1-x^2}$ are functionally dependent. Write the functional relation between u and v.
11. If u=x+y+z , $v=x^2+y^2+z^2$ -2xy-2yz-2xz, $w= x^3+y^3+z^3$-3xyz verify whether u, v and w are functionally dependent. If, so, find the relation between them.
12. Examine whether u= $\frac{x-y}{x+y}$, v= $\frac{x+y}{x}$are functionally dependent. If so, find the relation between them.
13. If u x+2y-z, v = 2x-y+z, w=x-y-z find the Jacobian $\frac{\partial(u,v,w)}{\partial(x,y,z)}$
14. If u = x+y+z, $v = x^2+y^2+z^2$, $w = x^3 + y^3 +z^3$ -3xyz find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$
15. If u = x+y, v = x-y prove that JJ^1 =1
16. If x = r$\sin\theta\cos\phi$, y = r$\sin\theta\sin\phi$, z = r$\cos\theta$ find the jacobian J
17. If u = $\frac{x^2+y^2}{x^2}$, v = $\frac{2xy}{x^2+y^2}$ find the jacobian J
18. If u = x+y+z, v = x-y+z, w = x+y-z verify that JJ^1 =1

MAXIMA AND MINIMA

1. Find the extremum values of the function $f(x,y) = x^3+y^3$-63(x+y) +12xy
2. Find the maximum and minimum values of $x^3+3xy^2-3x^2-3y^2+4$
3. Find the extremum values of the function $f(x,y) = x^2 y^2$ -5x^2-8xy-5y^2
4. Find the extremum values of the function $f(x,y) = x^4 + y^4$ -2x^2 + 4xy - 2y^2
5. Find the stationary points f(x, y) = sinx.siny.sin(x+y) where 0<x<π ,0<y<π and find the extremum of f.
6. Examine for minimum and maximum values of f(x, y) = sinx + siny + sin(x+y)

MAXIMA AND MINIMA (LAGRANGE'S METHOD)

1. Find a point on the plane 3x+2y+z-12=0, which is nearest to the origin.
2. Find a point on the plane x+2y+3z = 4, which is nearest to the origin
3. Find the minimum value of f = x^2+ y^2+ z^2 given x+y+z=3a
4. Find the maximum value of f = $x^2y^3z^4$ if 2x+3y+4z = a.
5. Find the maximum value of f=$x^2y^3z^4$, subject to the condition x+y+z=5
6. Find the minimum values of f=x^2yz^3, subject to the condition 2x+y+3z=a
7. Find the minimum value of $x^2 + y^2 + z^2$ when ax +b y+ cz =p
8. Given that x+y+z = a, find the maximum value of f = $x^my^nz^p$
9. Find the extremum value of the function f = x+y+z subject to the condition $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$
10. A wire of length b is cut into two parts, which are bent in the form of a square and a circle, respectively. Find the least value of the sum of the areas so found.
11. Find the shortest distance from the point (1,0) to the parabola y^2=4x
12. Find the points on the surface z^2 = xy+1 that are nearest to the origin
13. Find the points on the surface xyz^2 = 2 that are nearest to the origin
14. Divide 24 into three parts such that the continued product of the first, the square of the second and the cube of the third is maximum
15. The sum of three numbers is constant. Prove that their product is maximum when they are equal.

16. A rectangular box open at the top is to have a volume of 32 cubic ft. Find the dimensions of the box requiring the least material for its construction.
17. Show that the rectangular solid of maximum volume that can be inscribed in Sphere $x^2+y^2+z^2=a^2$ is a cube.
18. Find the volume of the greatest rectangular parallelepiped that can be inscribed in ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$
19. Divide 120 into three parts so that the sum of their products taken two at a time shall be maximum
20. The temperature T at any point (x,y,z)in space is $T = 400xyz^2$. Find the highest temperature of the surface of the unit sphere $x^2 + y^2 + z^2 = 1$

EULERS THEOREM

1. Verify Euler's Theorem for $u = x^2 tan^{-1}\left(\frac{y}{x}\right) + y^2 tan^{-1}(\frac{x}{y})$ and also prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2+y^2}$
2. If $u(x,y) = \log\left(\frac{x^4+y^4}{x+y}\right)$,show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$
3. If $u = sin^{-1}\left(\frac{x^2+y^2}{x+y}\right), x + y \neq 0$,prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan(u)$
4. Find $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial y \partial x} + y^2\frac{\partial^2 u}{\partial y^2}$, if $u = tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$
5. If u = $\frac{x^3y^3z^3}{x^3+y^3+z^3}$ +log($\frac{xy+yz+zx}{x^2+y^2+z^2}$) find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$
6. If u = $\sin^{-1}(\frac{x+y}{\sqrt{x}+\sqrt{y}})$ Prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}$ tanu and $x^2\frac{\partial^2 u}{\partial x^2}$ +2xy $\frac{\partial^2 u}{\partial x \partial y}$ + $y^2\frac{\partial^2 u}{\partial y^2} = \frac{-sinu\, cos2u}{4co^3 u}$
7. If u = $\cot^{-1}(\frac{x+y}{\sqrt{x}+\sqrt{y}})$ show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + \frac{1}{4}$ sin2u=0

MODULE-V

MULTIPLE INTEGRALS

SHORT ANSWER QUESTIONS

1. Evaluate $\int_0^{\pi}\int_0^{asin\theta} r dr d\theta$
2. Write limits of Spherical coordinates r,$\theta, \emptyset$ in (i) For the positive octant of the sphere (ii) for the hemisphere (iii) For a complete sphere
3. Evaluate $\int_0^1\int_x^{\sqrt{x}} x^2y^2(x+y)dy\,dx$
4. Evaluate $\int_0^1\int_y^1\int_0^{1-x} x\,dz\,dx\;dy$
5. Change the order of integration of $\int_0^b\int_y^{\frac{a\sqrt{b^2-y^2}}{b}} x\,y\,dx\;dy$
6. Evaluate $\int_0^{\pi/4}\int_0^{asin\theta}\frac{rdrd\theta}{\sqrt{a^2-r^2}}$
7. Evaluate $\iint x^2y^3\,dxdy$ over the rectangle $0\le x\le 1$ and $0\le y\le 3$
8. Evaluate $\int_0^1\int_y^1\int_0^{1-x} x\,dzdxdy$
9. Evaluate $\int_0^1\int_{y^2}^1\int_0^{1-x} x\,dz\,dx\,dy$
10. Transform the double integral $\int_0^a\int_0^{\sqrt{a^2-x^2}} y\,\sqrt{x^2+y^2}\;dxdy$ into polar coordinates

LONG ANSWER QUESTIONS

DOUBLE INTEGRALS

1. Evaluate $\iint_R xy\,dxdy$ where R is the region bounded by the X-axis, the ordinate x=2a and the curve $x^2 = 4ay$
2. Evaluate $\iint\frac{dxdy}{x^4+y^2}$, over the region bounded by the $y\ge x^2$, $x\ge 1$
3. Evaluate $\iint(x^2+y^2)dxdy$ in the positive quadrant for which x+y≤1
4. Evaluate $\iint_R e^{x^2}\,dxdy$ where R is the region R is given by R: $2y\le x\le 2\le and\ 0\le y\le 1$
5. Evaluate $\iint e^{y^2}dx\,dy$ over the region bounded by the triangle with vertices (0,0), (2,1), (0,1)
6. Evaluate $\iint_R (x+y)^2\,dx\,dy$ where R is the region bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$
7. Evaluate $\iint_R y\,dy\,dx$ where R is the region in the first quadrant bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$
8. Evaluate $\iint_R xy\,dxdy$ where R is the region bounded by the line x+2y=2 and lying in the first octant
9. Evaluate $\iint_R y\,dx\,dy$ where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2=4y$

AREA OF THE REGION

1. Find the area bounded by the parabolas $y^2 = 4ax$ and $x^2 = 4ay$
2. Find the area bounded by the y-axis, line y=2x and y=4
3. Find the area bounded by the parabola $y^2 = 4ax$ and the line 2x-3y+4-0
4. Find the area enclosed by the parabola $y^2=4ax$ and the lines x + y=3a, y=0 in the first quadrant
5. Find the area bounded by the curves y = 2-x and $y^2 = 2(2-x)$

CHANGE THE ORDER OF INTEGRATION

1. Change the order of integration and evaluate $\int_0^\infty \int_0^x x\, e^{\frac{-x^2}{y}}$ dy dx
2. Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy\, dxdy$ and hence evaluate the double integral.
3. Change the order of integration and evaluate $\int_0^a \int_x^a (x^2 + y^2)$ dy dx
4. Change the order of integration and evaluate $\int_0^\pi \int_x^\pi \frac{siny}{y}$ dydx
5. Evaluate $\int_0^1 \int_{e^x}^e \frac{dydx}{logy}$ by changing the order of integration
6. By changing the order of integration, evaluate $\int_0^3 \int_1^{\sqrt{4-y}} (x + y)$ dxdy
7. Evaluate by changing the order of integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}$ dy dx
8. By changing the order of integration, evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} xy\, dy\, dx$
9. By changing the order of integration, evaluate $\int_3^5 \int_0^{4/x} xy\, dy\, dx$
10. Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2)\ dydx$ by changing the order of integration
11. By changing the order of integration, evaluate $\int_0^1 \int_{\sqrt{y}}^{2-y} xy\, dx\, dy$
12. By changing the order of integration, evaluate $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{y^2+x^2}}$ dy dx
13. By changing the order of integration, evaluate $\int_0^a \int_{\sqrt{ax}}^a \frac{y^2}{\sqrt{y^4-a^2x^2}}$ dy dx

POLAR COORDINATORS

1. Calculate $\iint r^3$ drdθ over the area included between the circles r=2sinθ and r=4sinθ
2. Evaluate $\iint \frac{r\, dr\, d\theta}{\sqrt{a^2+r^2}}$ over one loop of the lemniscates $r^2 = a^2 \cos 2\theta$
3. Show that $\iint_R r^2 \sin\theta\, drd\theta = \frac{2a^2}{3}$ where R is the semi circle r= 2a sinθ and above the initial line
4. Evaluate $\iint r^4 \sin^3\theta$ drdθ over the region of circle r = 2cosθ
5. Evaluate $\iint rsin\Theta$ drdΘ over the cardioid r=a(1-cosΘ) above the initial line

AREA OF THE REGION

1. Find the area bounded between the circles r = 2cosθ and r = 4sinθ
2. Find the area common to the cardioids r = a(1+cosθ) and r=a(1-cosθ)
3. Find by double integration the area lying inside the circle r = asinθ and outside the cardioid r = a(1-cosθ)
4. Find the area of the crescent bounded by the circles r=$\sqrt{3}$ and r= 2cosθ
5. Find the area which is inside the cardioids r = a(1+cosθ)and outside the circle r = a
6. Find the area that lies inside the circle r=3acosθ and outside the cardioid r=a(1+cosθ)
7. Find the whole area of the lemniscates $r^2 = a^2 \cos 2\theta$
8. Using the double integral, find the area of the cardioid r=a(1-cosΘ)
9. Using the double integral, find the area of the cardioids r=a(1+cosΘ)
10. Calculate $\iint r^3$ drdθ over the area included between the circles r=2sinθ and r=4sinθ

CHANGE OF VARIABLES (CARTESIAN TO POLAR COORDINATES)

1. Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}$ dx dy. Deduce that $\int_0^\infty e^{-x^2}$dx = $\frac{\sqrt{\pi}}{2}$
2. Evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2}$ dxdy by transforming into polar coordinates
3. Evaluate the integral $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} x^2$ dy dx by changing into polar coordinates
4. Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{x^2+y^2}$ dy dx by transforming into polar coordinates
5. Evaluate $\int_{-\infty}^\infty \int_{-\infty}^\infty \frac{dxdy}{(1+x^2+y^2)^{3/2}}$
6. Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} y\sqrt{x^2+y^2}$ dx dy transforming into polar coordinates
7. Evaluate $\iint \frac{x^2y^2}{x^2+y^2} dxdy$ over the region bounded by the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ (b>a)

TRIPLE INTEGRALS

1. Evaluate $\int_0^a \int_0^x \int_0^{1-x-y} \frac{1}{(x+y+z)^3}$ dxdydz
2. Evaluate $\iiint_v (xy + yz + zx)dxdydz$ where V is the region of space bounded by x=0, y=0, z=0, x=1, y=2, z=3.
3. Evaluate $\int_0^1 \int_0^x \int_0^{x+y} e^{x+y+z}$ dzdydx
4. Evaluate $\iiint xyz\, dxdydz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$
5. Evaluate $\iiint \frac{dxdydz}{\sqrt{a^2-x^2-y^2-z^2}}$ over the bounded by the sphere $x^2 + y^2 + z^2 = a^2$
6. Evaluate $\iiint dx\, dy\, dz$ where V is the finite region of space formed by the planes x=0, y=0, z=0,2x+3y+4z=12
7. If R is the region bounded by the planes x=0,y=0, z=1 and the cylinder $x^2+ y^2 =1$ evaluate the integral $\iiint_R xyz\, dxdydz$ by changing into cylindrical polar coordinates
8. Evaluate $\iiint_v \frac{dxdydz}{(x+y+z+1)^3}$ taken over the volume bounded by the planes x=0,y=0,z=0 and the plane x + y + z=1
9. Evaluate $\iiint (x + y + z)dxdydz$ over the tetrahedron bounded by the coordinate planes and plane x+y+z=1
10. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{1}{(1+x+y+z)^3}$ dxdydz
11. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dzdydx$
12. Evaluate $\iiint (x + y + z)dxdydz$ over the tetrahedron bounded by the coordinate planes and the plane x+y+z=1
13. Evaluate $\int_1^e \int_1^{logy} \int_1^{e^x} logz\, dzdxdy$
14. Evaluate $\int_0^\infty \int_0^\infty \int_0^\infty \frac{dxdydz}{(x^2+y^2+z^2)^2}$

VOLUME INTEGRALS

1. Find the volume of the tetrahedron bounded by the planes x=0,y=0,z=0 and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$
2. Using a triple integral, find the volume of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$
3. Calculate the volume of the solid bounded by the planes x=0,y=0,x+y+z=a &z=0
4. Evauate $\iiint_v dxdydz$ where V is the finite region of the space formed by the planes x=0,y=0,z=0 and 2x+3y+4z=12

CHANGE OF VARIABLES (CARTESIAN TO SPHERICAL AND CYLINDRICAL CORDINATES)

1. Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}}\frac{1}{\sqrt{1-x^2-y^2}}\,dzdydx$ by changing to spherical polar coordinates
2. Using cylindrical coordinates, find the volume of the cylinder with base radius a and h
3. Change the integral $\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{y^2+x^2}}^1\frac{dxdydz}{\sqrt{x^2+y^2+z^2}}$ to spherical polar coordinates and evaluate it.
4. Evaluate $\iiint(x^2+y^2+z^2)$ dxdydz taken over the volume enclosed by the sphere $x^2+y^2+z^2=1$ by transforming into spherical polar coordinates.
5. Using triple integrals, find the volume of the sphere $x^2+y^2+z^2=a^2$
6. Using spherical polar coordinates $\iiint\frac{dxdydz}{x^2+y^2+z^2}$taken over the volume enclosed by the sphere $x^2+y^2+z^2=a^2$
7. If R is the region of the 1st octant bounded by the sphere $x^2+y^2+z^2=a^2$, Evaluate $\iiint_R(x+y+z)dxdydz$ by changing to spherical polar coordinates
8. Evaluate $\iiint\frac{dxdydz}{(x^2+y^2+z^2)^{1/2}}$ over the region bounded by the spheres $x^2+y^2+z^2=a^2$ and $x^2+y^2+z^2=b^2$ Where a>b>0.